

AI-Powered Multi-Agent PDF Analysis System

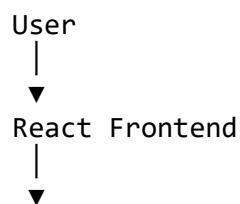
An Agentic AI-based PDF Analysis Platform built using **LangGraph**, **LangChain**, **FastAPI**, **ReactJS**, **FAISS**, and **Ollama (Qwen2.5:7B)**.

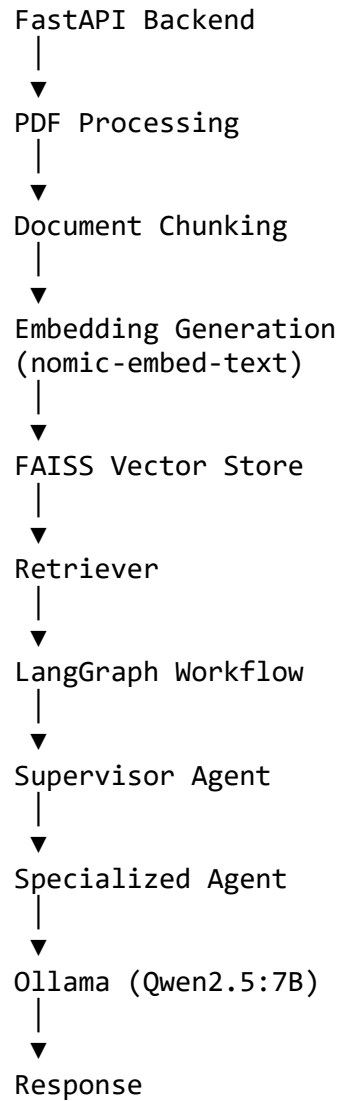
The system allows users to upload PDF documents and interact with them using natural language. It leverages a multi-agent architecture to perform document summarization, explanation, comparison, methodology extraction, citation generation, financial analysis, KPI extraction, trend analysis, risk assessment, and more.

Features

-  User Authentication using JWT
 -  PDF Upload and Processing
 -  Retrieval-Augmented Generation (RAG)
 -  Multi-Agent AI Architecture
 -  Semantic Search using FAISS
 -  Context-Aware Question Answering
 -  Document Summarization
 -  Financial Report Analysis
 -  KPI & Trend Analysis
 -  Risk Assessment
 -  Investor Insights
 -  Methodology Extraction
 -  Citation Generation
 -  Quiz Generation
 -  Fully Local LLM Deployment using Ollama
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System Architecture





Tech Stack

Frontend

- React.js
- JavaScript (ES6+)
- HTML5
- CSS3
- React Hooks
- Fetch API

Backend

- Python
- FastAPI
- SQLite
- JWT Authentication

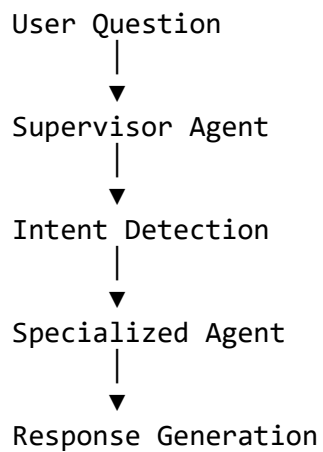
AI & Machine Learning

- LangGraph
 - LangChain
 - Ollama
 - Qwen2.5:7B
 - nomic-embed-text
 - FAISS
-

Agentic AI Architecture

The project follows a Multi-Agent Architecture using LangGraph.

Instead of sending every request to a single prompt, a Supervisor Agent analyzes the user's query and routes it to the most suitable specialized agent.



Agents Used

Supervisor Agent

Responsible for:

- Query understanding
 - Intent classification
 - Agent selection
 - Workflow orchestration
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Summary Agent

Generates concise summaries of uploaded documents.

Example:

Summarize the document

Explain Agent

Provides detailed explanations of concepts and topics.

Example:

Explain Disk Scheduling

Comparison Agent

Compares concepts, algorithms, technologies, or entities.

Example:

Compare FCFS and SSTF

Methodology Agent

Extracts methodologies, workflows, and procedures from research papers and technical documents.

Citation Agent

Identifies references and citations from documents.

Quiz Agent

Generates:

- MCQs
 - Short Answer Questions
 - Learning Assessments
-

Financial Summary Agent

Creates executive summaries for financial reports.

KPI Agent

Extracts key business metrics such as:

- Revenue
 - Profit
 - EBITDA
 - EPS
 - Operating Margin
-

Trend Agent

Analyzes:

- Revenue Growth
 - Profit Trends
 - Historical Performance
-

Risk Agent

Identifies:

- Business Risks
 - Operational Risks
 - Financial Risks
-

Investor Agent

Provides investor-focused insights and observations.

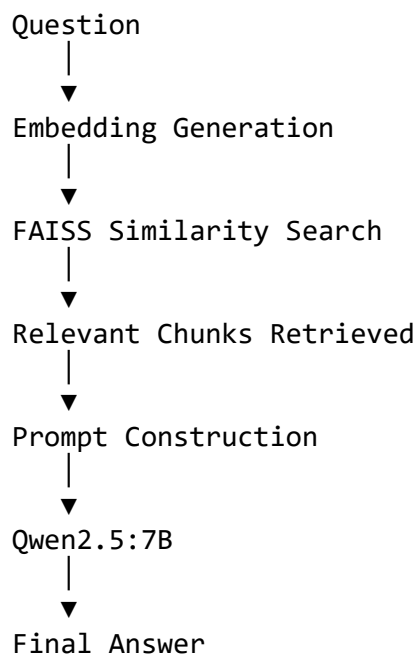
Balance Sheet Agent

Analyzes:

- Assets
 - Liabilities
 - Equity
 - Debt
 - Liquidity
-

Retrieval-Augmented Generation (RAG)

The application follows a RAG architecture to ensure document-grounded responses.

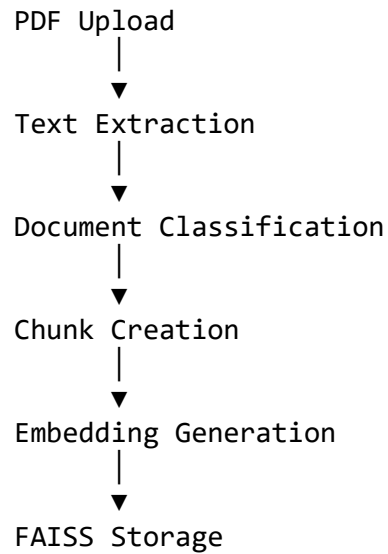


Benefits

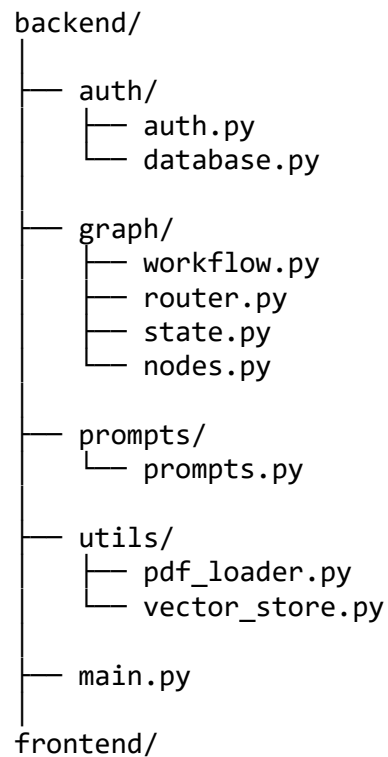
- Reduced hallucinations
- Better accuracy
- Context-aware responses

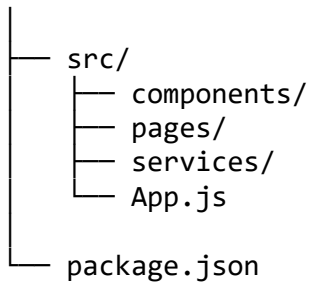
- Improved document understanding
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PDF Processing Pipeline



Project Structure





Installation

Clone Repository

```
git clone https://github.com/sahsanidhristi/Agentic-AI-pdf-Chatbot-with-Ollama-and-react.git
```

```
cd Agentic-AI-pdf-Chatbot-with-Ollama-and-react
```

Backend Setup

Create Virtual Environment

```
python -m venv venv
```

Activate Environment

Windows:

```
Venv/Scripts/activate
```

Linux/Mac:

```
source venv/bin/activate
```

Install Dependencies

```
pip install -r requirements.txt
```

Ollama Setup

Install Ollama

```
ollama pull qwen2.5:7b
```

```
ollama pull nomic-embed-text
```


Start Ollama

```
ollama serve
```

Run Backend

```
uvicorn main:app --reload
```

Backend URL:

<http://localhost:8000>

Frontend Setup

Navigate to frontend folder

```
cd frontend
```

Install packages

```
npm install
```

Run application

```
npm start
```

Frontend URL:

<http://localhost:3000>

Authentication Flow

Register



Login



JWT Token Generation



Protected API Access



Advantages

- Multi-Agent Architecture
 - Context-Aware Responses
 - Local LLM Deployment
 - No API Cost
 - Privacy Preserving
 - Fast Semantic Retrieval
 - Modular Design
 - Easy Agent Expansion
 - Financial and Educational Analysis Support
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Future Enhancements

- Multi-PDF Chat
 - OCR Support
 - Voice Input
 - Speech Output
 - Multi-Language Support
 - Cloud Deployment
 - Agent Collaboration
 - Graph-Based Knowledge Retrieval
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Author

Developed as an Agentic AI-powered document intelligence platform using LangGraph, LangChain, FAISS, FastAPI, ReactJS, and Ollama.



License

This project is intended for educational, research, and demonstration purposes.